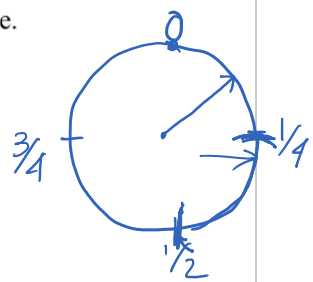
04/20/2022

Wednesday, April 20, 2022 10:53 AM

M168_Notes_8.5

8.5 The Normal Distribution

Continuous random variables can take on any value.



Let t = time in seconds to run a race

Let w = weight of kitten in kg

Let L = length of a week-old bean plant

Let X = value where pointer lands.

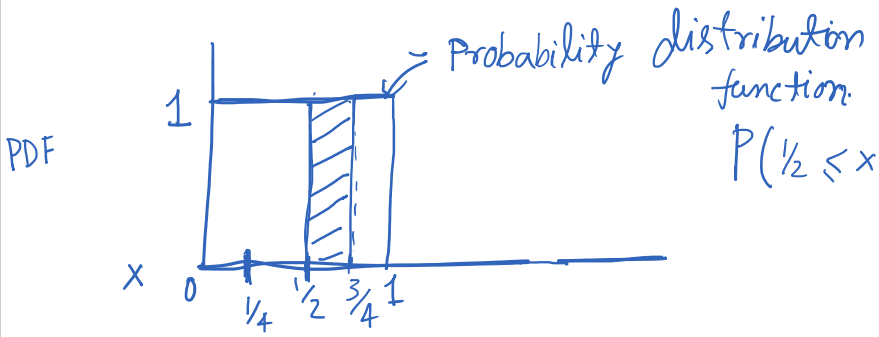
$0 \le X < 1$ and $P(0 \le X < 1) = 1$

What is $P(X = \frac{1}{2})$? $= 0$

What is $P(0 \le X < \frac{1}{4})$? $= \frac{1}{4}$

What is $P(0.75 \le X \le 0.80)$? $= 0.05$

Define a PROBABILITY DENSITY FUNCTION



$P(\frac{1}{2} \le x \le \frac{3}{4})$ = Area of the shaded region.

$= \frac{1}{4}$

Discrete finite variables - graph the probability as a histogram.

Each rectangles have a base of width 1 (centered on X) and the height was P(X).

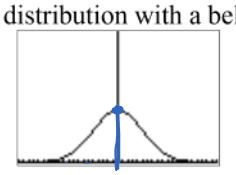
So the area, length times height was the probability that X occurred.

AREA above our X value will be the probability that get that X ← value.

If we want to find the probability of a range of X values, we would add up the areas over the range of X values.

When we graph the probability distribution for a continuous variable we find a smooth curve.

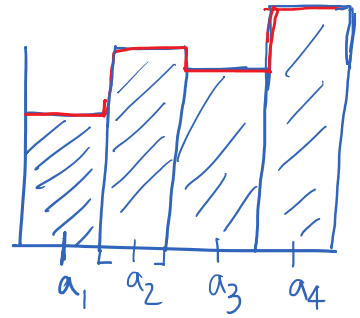
Many natural and social phenomena produce a continuous distribution with a bell-shaped curve.



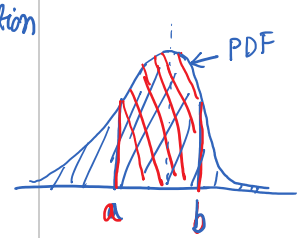
μ ← Expectation
μ = Median

Every bell-shaped (NORMAL) curve has the following properties:

- Its peak occurs directly above the mean μ
- The curve is symmetric about a vertical line through μ. The curve never touches the x-axis. It extends indefinitely in both directions.



$P(a \leq X \leq b)$



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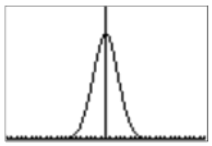
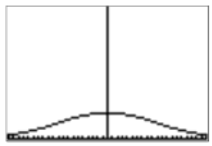
$$\int_a^b P(x)\,dx = P(a \leq x \leq b)$$

- The area between the curve and the x-axis is always 1 (total probability is 1).

The shape of the curve is completely determined by : and Φ,

$$P(x)=\frac{1}{\sqrt{2ps}}e^{-\frac{(x-m)^2}{2s^2}}$$





$<\mu$

$\sigma>$

The probability that a data value will fall between x=a and x=b is given by the area under the curve between x=a and x=b.

The standard normal curve has $\boxed{\mu=0}$ and $\boxed{\sigma=1}$. ~~Use Z, NOT X~~.

Example

On a standard normal curve, what is the probability that a data value is between -1 and 1, P(-1<z<1)?

68.27%

normalcdf (a, b)

Example

Suppose that X is a normal random variable with μ=50 and σ=10.

a) What is the probability that X<30? normalcdf $(-\infty, 30, 50, 10) = 0.0228$

b) What is the probability that 35<X<65?

0.8663

Remember Chebychev's theorem?

$P(35 \leq x \leq 65) \geq 1 - \frac{1}{(1.5)^2} \approx 0.56$

$P(\mu - K\sigma \leq x \leq \mu + K\sigma) \geq 1 - \frac{1}{K^2}$

Example: An instructor wants to "curve" the grades in his class. The class mean at the end of the semester is 73 with a standard deviation of 12. He decides that the top 12% of the class should get an A, the next 24% should get a B, the next 36% a C, the next 18% a D and the last 10% of the class will get an F. What are the cutoffs for the grades?

Assumption: Scores follows normal distribution

$\mu = 73, \ \sigma = 12$

A cutoff: InvNorm$(0.88, 73, 12) = 87.1$

B cutoff: InvNorm$(0.64, 73, 12) = 77.3$

C cutoff: InvNorm$(0.28, 73, 12) = 66.0$

D cutoff: InvNorm$(0.1, 73, 12) = 57.62$